# Manual

## Archiving Machine Data
## MDE-ARC 8.1

Version 1.0.5003

Last changed on: 19.06.2020

## Copyright

# Contents

# 1 Archiving of Machine data

## Purpose

The machine data archiving function package not only provides the ability to access archive data, but also to export recorded data and to then upload the data back into the system as needed.

## Implementation considerations

You use the machine data archiving function package if:

- You would like to have access to data already moved to archive tables within the function package's machine data entry feature,
- Because of legal requirements or customer demands, you need to ensure that the data entered is stored long-term.
- Because of legal requirements or customer demands, you need to import data that was already exported back into the system to be evaluated / analyzed again.

## Integration

Machine data archiving accesses recorded data in the respective archive tables of the machine data collection function package. On the other hand, master data from the machine data collection function package are not archived, but remain in the on-line tables.

## Features

- Direct access to archive tables from the application
  - Direct access to archived machine data from the applications
    - Machine history
    - Machine time profile
    - Efficiency report
    - Performance profile
    - OEE report
    - Status report (workplace/machine-related)
    - Status report
    - Status profile
    - Status class report
    - Status class profile
    - RPA report
    - RPA profile

        - Status analysis
- Transport function
    - Functions for transporting data from the on-line tables into archive tables.
- Export function
    - Function for transferring (exporting) data from the archive tables to external file systems for the purpose of long-term storage of the recorded machine data.
- Import function
    - Functions for importing exported data into the archive tables in order to evaluate them using the applications listed above.

# 2 MDE-specific Configuration

## Overview

The data are stored as standard for 35 days in the MDE module before they are moved to long-term storage.

Various MDE evaluations allow data to be accessed which are older than 35 days. For this, the BDE postings are stored in a special medium-term or archive area. Access to these data is essentially automatic as soon as the selection period exceeds the short-term data range. In a few applications there is the option "Consider long-term data" in the selection area via which these data can be accessed.

Data included in the MDE archiving are:

- MDE log records (documents)
- MDE events
- Additional information (e.g. comments)
- LOGGING entries for the machine

## Configuration

The retention period of the data in the individual data areas can be configured via the HYDRA data management.

The data management does not define the retention period for data of the “cycle progression” application. The relevant configuration is described in the document dealing with the cycle progression application.

During the transfer of the data to the archive tables, those data are transferred whose "retention period" (in days/months/years; see values in parentheses in the following table) has been exceeded. If the relevant archiving license for the MDE is not available, the data are deleted after the set retention period.

| Product | Object | Object designation | Transfer | Interval in as-delivered condition |
|---|---|---|---|---|
| MDE | MDEPRO | Log records / documents | Online data<br>→ medium-term archive | 35 days 1) |

| Product | Object | Object designation | Transfer | Interval in as-delivered condition |
|---|---|---|---|---|
| MDE | A_MDEPRO | Long-term archiving of log records / documents | Medium-term archive<br>→ Long-term archive | 3 years 1) |
| MDE | EREIGMDE | Events | Online data<br>→ medium-term archive | 35 days 2) |
| MDE | A_ EREIGMDE | Long-term archiving of events | Medium-term archive<br>→ Long-term archive | 3 years 2) |
| MDE | LOG | HYDRA logging data | Online data<br>→ medium-term archive | 35 days |
| MDE | A_LOG | Long-term archiving of HYDRA logging data | Medium-term archive<br>→ Long-term archive | 3 years |
| MDE | RES_STATUS | Parallel MDE resource status | Online data<br>→ medium-term archive | 70 days |
| MDE | A_ RES_STATUS | Long-term archiving of parallel MDE resource status | Medium-term archive<br>→ Long-term archive | 3 years |

**Please note:**

1) Please note that the postings relating to workplaces/machines only allow for data of the online data area to be selected and edited.
2) Please note that the event maintenance only allows for data of the online data area to be selected and edited.

# 3 Data Management

## Overview

| | |
|---|---|
| Menu | System administration → Archiving → Data management |
| Transaction code | arccfg |
| Function authorization | arccfg.* |

## Usage

You may use this application in order to view or edit the centrally managed settings for archiving.

## Integration

The settings for archiving/for data management are made centrally from all components/functions in the application.

## Field descriptions

**Product**

In the Product field, the HYDRA module for which a rule is to be defined is entered.

**Object**

The Object defines what is to be provided .

**Retention period**

In the "retention period" fields, you define how long the data should be available before being archived.

**Unit**

The unit for the retention period is indicated in days, months or years.

**Last retention date**

This field is computed and subsequently indicates the last day on which data are still available in this object.

**Action**

It is possible to select from three process variants for an object:

| | |
|---|---|
| D = delete | Object is deleted |
| M = move (archive) | Object will be transferred to the next division |
| X = Export | Object is unloaded (XUNLOAD format), subsequently deleted |

**Target object**

At present, this automatically results from the detail configuration.

**Condition**

The characters in this field are added to the database command controlling the action, whereby this condition is linked to the conditions of the original command there and is set in parenthesis.

**Last run (date/time)**

Indicates when this rule was applied last.

**License**

Indicates which license is required for archiving. Multiple licenses may be entered here (separated by a space). If none of these licenses is licensed on the system, the data are either deleted (action D) or unloaded in files (action M or X) according to the relevant action.

**Path**

Optional path for generating the file export (Unload). At present, only local drives are supported on the HYDRA server in the archive. If no path is set, file exports are filed as follows:

**<HYDRADIR>/<SYSTEM>/custom/archive/<YYYY-MM-DD>/<PRODUCT>/**

HYDRADIR … HYDRA directory
SYSTEM … System number
YYYY-MM-DD … Archiving date (YYYY … Year, MM … Month, DD … Day)
PRODUCT … Product from archiving configuration

**Management table**

Table name of management table where the archiving logs are stored.

**Retention period management table**

In the fields used for the Retention period, you define how long the logs should be available in the management table before being deleted.

**Unit**

The unit for the retention period is indicated in days, months or years.

**Archiving step**

Indicates whether this archiving executes archiving function I or II. In archiving function I, the data are transferred from the online inventory to the archive tables and/or deleted. Setting M (medium-term archive) In archiving function II, the data are transferred from the archive table to the file export and/or deleted. Setting L (long-term archive).

**Configuration**

Control indicator whether or not the configuration is active. Possible values: Y/N.

**Archiving type**

Identifier for time or object-related archiving. Supported modes: O = Object-related archiving (i.e. data are archived for each object individually). Z = Time-related archiving (i.e. data are collected/archived without any object reference). Please note: Time and object-related archiving differ from each other significantly in the archiving performance (runtime). Object-related archiving of mass data is not recommendable.

**Priority**

Integral value greater than 0. Indicates the execution sequence when several objects are defined to a module. Execution starts from the configuration with the lowest value.

**Master table**

Table including the archiving data. The extensions entered in the Condition field refer to this table.

**Date column**

Date column in the master table. Is used for evaluating the data to be archived with regard to the retention period.

**Key 1**

Clear key column in the master table; may be used to identify the data to be archived. Up to 5 key columns may be defined for object-related archiving. Time-related archiving only supports one primary key in Key 1.

**Keys 2 – 5**

Additional optional key columns for object-related archiving.

**Comment**

Comment line for describing the archiving configuration.

When copying the archiving configuration, only the data management configuration is copied at present, but no any defined data records from the object details. At present, the data records from the object details must be copied to the new configuration by the user in the appropriate screen.

| **Module** | **Version** | **Description of archiving** |
|---|---|---|
| BDE | 8.1 | here |
| CAQ | 8.1 | here |
| MDE | 8.1 | here |
| MPL | 8.1 | here |

# 4 Reload Manager

## Summary

| | |
|---|---|
| Menu | System Administration → Archiving → Reload Manager |
| Transaction code | arcrld |
| Function authorization | arcrld<br>arcrld.export<br>arcrcl.import |

## Utilization

The customer is responsible for backing up data in a data archive and for restoring data. When data is backed up it is transferred from the long-term data area to the archive data area and, as a result, it is stored in a separate file system. The customer can start the backup process using the Reload Manager.

When data, which is filed in the archive data area, is restored, it is transferred back (copied) to the long-term data area. The customer has to do this and bears the responsibility.

To be able to evalaute restored data using standard evaluations/reports, the data has to be transferred back (copied) to the reload data area.

For this reason, the Reload Manager enables the following functions:

1. Moving of exported data into the customer archive (the customer archive path has to be defined within the HYDRA path settings).
2. Loading of exported data into the reload area for evaluations

## Integration

The Reload Manager is a central function that is used by many components or functions.

## Selection Criteria

The application provides the following selection criteria:

**Module**

Reference to the product group which archived data belongs to.

**Object type**

Object type of the archived data.

The selection options “module” and “object type” are mandatory fields.

## Toolbar

### Export

The “EXPORT” button allows for the entries selected in the Reload Manager to be exported to the customer directory (all data records can be selected using the context menu of the right mouse button). Once the function has been started, the input dialog that opens requires the customer archive path to be entered.

### Import

The “IMPORT” button allows for the selected entries to be loaded into the reload area, in order for the data to be again available for HYDRA evaluations/reports. Once the function has been started, the data loading mode and an optional path are requested. By indicating the path, it is possible to specify another storage location for the files, provided that the archived files have not been moved to the customer-specific archive using the Reload Manager.

There are three different ways to deal with reload data after they have been reloaded to prevent the reload data set from increasing excessively (→ no slow evaluations/reports):

The following modes are distinguished:

- Cyclic
  Data is loaded to the reload area in the “cyclic” mode. Loaded data is automatically removed from the corresponding reload tables, once the retention period specified within archive settings has expired.
  Please compare the configuration entry HYD / RELOADMANAGER of the data management function. When demo settings are used, data is removed from the reload area after 14 days.

  When data is imported, the HYD_REL_MANAGEMENT.DELETE_DATE field is calculated subject to the configuration and entered in the reload management table.
- User-specific
  Data is loaded to the reload area in the “user-specific” mode. The loaded data is automatically removed from the corresponding reload area, once the time specified for the user within the user-specific settings has expired. Exception: In case several users loaded identical data into the reload area, the corresponding retention period is automatically set to the maximum date.
  **When data is imported, the HYD_REL_MANAGEMENT.DELETE_DATE field is computed subject to the configuration and entered in the reload management table.**

- Manual
  Data is loaded in the “manual” mode to the reload area. The customer is responsible for deleting data from corresponding reload tables. However, identical data cannot be loaded several times in the “manual” mode (data can be loaded in the “user-specific” mode though).

In order for data to be transferred to the reload data area, the HYDRA server must be able to access these files. Otherwise, loading is cancelled with an error message.